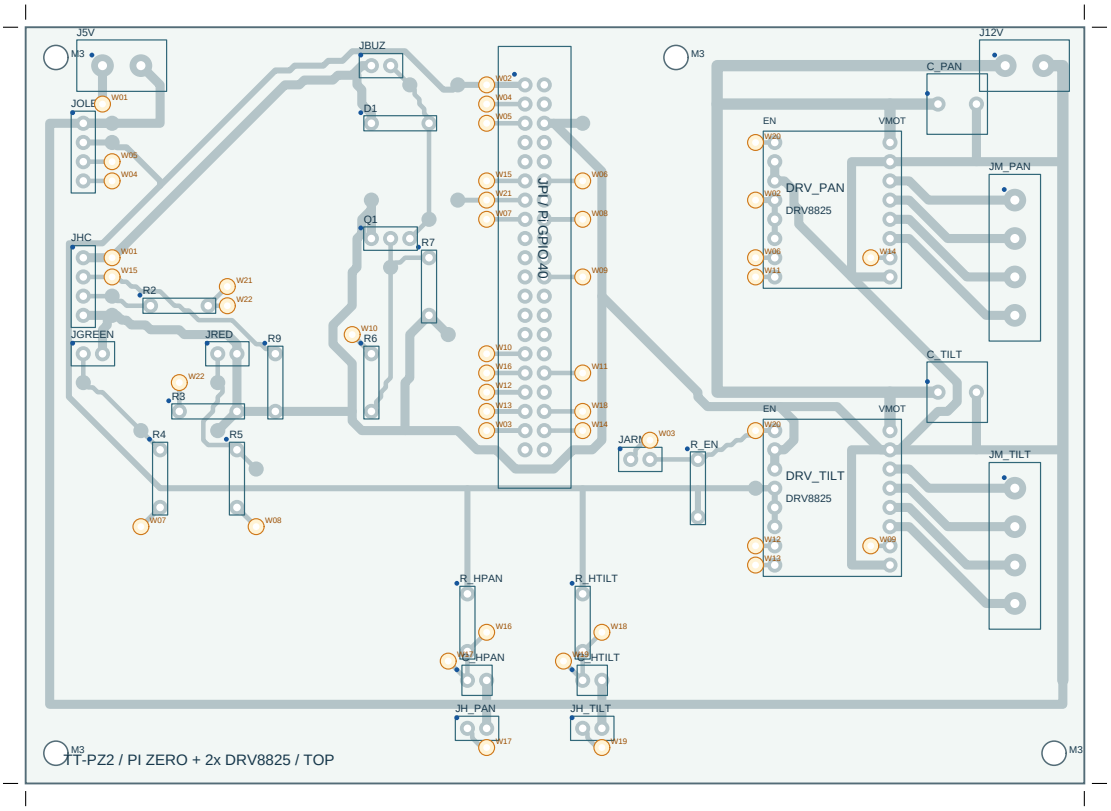
COMPONENT SIDE - 1:1 PLACEMENT

TT-PZ2 / 140 x 100 mm / exact scale at 100% print



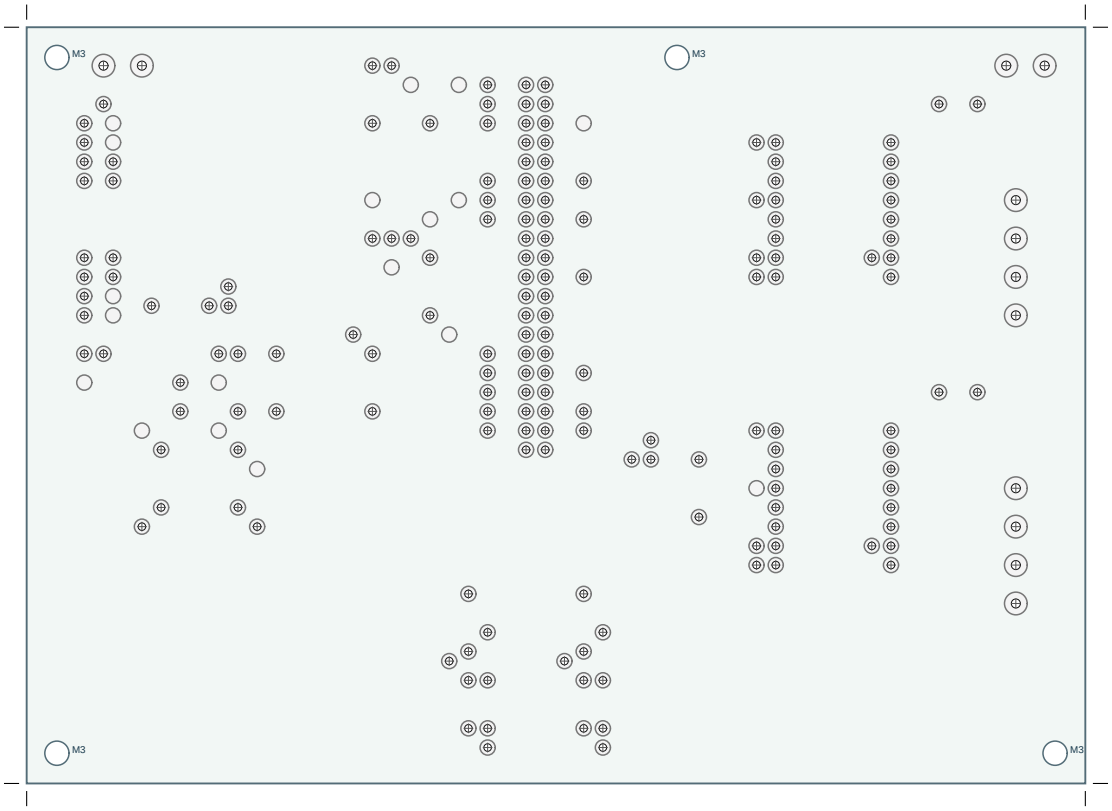
CALIBRATION: exactly 50 mm

Board outline 140 x 100 mm. A4 / Actual size / 100%.

All components on bare FR4 side. Copper is on the opposite side.
Blue dot = pin1. All copper pads round. JPI cable must be pin1-to-pin1.
DRV: EN upper-left, VMOT upper-right. HC TRIG direct3.3V; no buffer.
Capacitor pin1 positive. D1 pin1 cathode. Q1 pads E-B-C left-to-right.
Keep JARM open until STEP/DIR software and homing are ready.

DRILL GUIDE - COMPONENT SIDE

TT-PZ2 / 140 x 100 mm / exact scale at 100% print



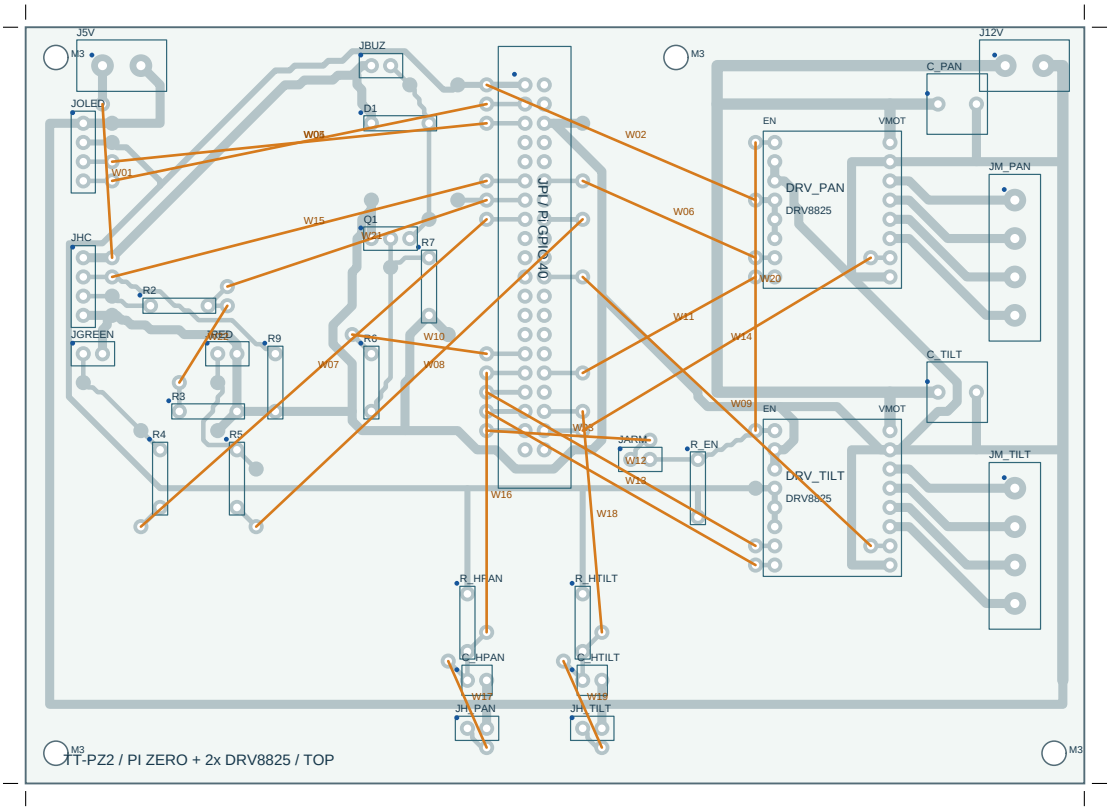
CALIBRATION: exactly 50 mm

Board outline 140 x 100 mm. A4 / Actual size / 100%.

Header / component / jumper holes: 1.0 mm.
Screw terminals: 1.2 mm. Mounts: 3.2 mm. Solid test pads: DO NOT DRILL.
Coordinates in pads.csv are millimetres from TOP-LEFT, component-side view.
Check footprint with the real part; form radial capacitor leads to 5.08 mm.
No plated holes required. Solder all pads on copper side.

INSULATED WIRE JUMPERS - TOP SIDE

TT-PZ2 / 140 x 100 mm / exact scale at 100% print



CALIBRATION: exactly 50 mm

Board outline 140 x 100 mm. A4 / Actual size / 100%.

Orange lines are insulated WIRE, not copper traces.
Each Wxx connects its two dedicated pads. Follow jumpers.csv endpoints.
Route around sockets/components as needed; cut length includes 20 mm slack.
AWG22 for GND / 5V / 12V; AWG26 for logic. Insulate every crossing.
Install wires before sockets; retain access to module trimpots and heatsinks.

WIRE SCHEDULE - COMPONENT SIDE

Pad coordinates from top-left. Dedicated pads already included in copper master.

REF	NET	PAD 1 (x,y mm)	PAD 2 (x,y mm)	CUT mm
W01	5V_AUX	11.43, 30.48	10.16, 10.16	40.4
W02	3V3	96.52, 22.86	60.96, 7.62	58.7
W03	ENABLE_GPIO	82.55, 54.61	60.96, 53.34	41.6
W04	SDA	11.43, 20.32	60.96, 10.16	70.6
W05	SCL	11.43, 17.78	60.96, 12.70	69.8
W06	PAN_STEP	96.52, 30.48	73.66, 20.32	45.0
W07	GREEN_GPIO	15.24, 66.04	60.96, 25.40	81.2
W08	RED_GPIO	30.48, 66.04	73.66, 25.40	79.3
W09	TILT_FAULT	111.76, 68.58	73.66, 33.02	72.1
W10	BUZZ_GPIO	43.18, 40.64	60.96, 43.18	38.0
W11	PAN_DIR	96.52, 33.02	73.66, 45.72	46.2
W12	TILT_STEP	96.52, 68.58	60.96, 48.26	61.0
W13	TILT_DIR	96.52, 71.12	60.96, 50.80	61.0
W14	PAN_FAULT	111.76, 30.48	73.66, 53.34	64.4
W15	TRIG_3V3	11.43, 33.02	60.96, 20.32	71.1
W16	HOME_PAN	60.96, 80.01	60.96, 45.72	54.3
W17	HOME_PAN	60.96, 95.25	55.88, 83.82	32.5
W18	HOME_TILT	76.20, 80.01	73.66, 50.80	49.3
W19	HOME_TILT	76.20, 95.25	71.12, 83.82	32.5
W20	DRV_EN	96.52, 53.34	96.52, 15.24	58.1
W21	ECHO_3V3	26.67, 34.29	60.96, 22.86	56.1
W22	ECHO_3V3	20.32, 46.99	26.67, 36.83	32.0